Systematic Literature Review: Corpus Linguistics in Indonesia

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Abstract. Native languages in many countries are currently in the risk of extinction, which happens in Indonesia. The lack of awareness of the younger generation and the globalization effects are mentioned as factors that play a role in the extinction. The fewer local language users, the more limited linguistic corpus will be. The languages in Indonesia itself also have the same problem mentioned in many literature, lacking of a linguistic corpus. This lack of a linguistic corpus causes difficulties in research, especially those which are based on artificial intelligence or natural language processing. This study conducted a systematic literature review on the corpus linguistics in Indonesia. The main purpose is to obtain information from various scientific articles ranging from the development to the use of the corpus linguistics in Indonesia. The results showed that the corpus mostly used primary sources such as social media, Wikipedia or previous similar research. There is no single article that discusses the development of the corpus of languages in Indonesia. Thus, research for the development of a corpus of languages in Indonesia is still wide open.

Keywords: Corpus linguistic · Native languages · Indonesia

1 Introduction

As of 2017, 668 languages from 2,468 regions in Indonesia have been identified by the Language Development and Development Agency, Ministry of Education and Culture [12]. Some of these local languages are still widely used by native speakers, such as Javanese and Sundanese. Although it is still used by many speakers, this does not mean that the local language is in are not about to extinct [19]. Almost all regional languages in Indonesia are going to extinct since the numbers of their speakers are decreasing [8].

The threat of various regional languages or mother tongues extinction does not only occur in Indonesia but also occurs in other countries. Research [24]

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C. Stephanidis et al. (Eds.): HCII 2022, CCIS 1580, pp. 370–377, 2022.
https://doi.org/10.1007/978-3-031-06417-3_50

states that around nine mother tongues are lost within one year. Today a third of the world's languages have fewer than 1,000 speakers and according to the United Nations Educational and Cultural Organization (UNESCO) world language atlas between 1950 and 2010 as many as 230 languages became extinct [25]. The loss of many mother tongues has attracted the attention of many linguists around the world, including UNESCO who designated February 21 as Mother Language Day as a proof that this issue is getting more crucial. The extinction of a language can result in the loss of sources of knowledge from people who use that language, such as architecture, medicine, agriculture, and shipping [30]. Many factors are said to be the cause of a language being no longer used by speakers, for example competition with a more dominant language. Globalization is also mentioned as one of the ten factors causing the extinction of regional languages in the [28] research. The ability to have an international language in the era of globalization is another reason why regional languages are increasingly being abandoned.

Lack of awareness of the local language as an intellectual property, especially by the younger generation, is another reason that causes regional languages to be abandoned [6]. The impact of globalization such as the existence of a majority and a minority, the lack of interest of the younger generation to learn regional languages which are their ancestral heritage are the factors causing the extinction of regional languages [16]. On the other hand, currently a lot of academics use language as a part of their research, such as sentiment analysis. The amount of digital data is currently a challenge in itself to be processed into a source of information. So the existence of a system work automatically finding opinions from the data becomes a necessity [16].

However, Indonesian language is categorized as a language that lacks of resources (especially labeled data set) that can be used for language processing technology [29]. A collection of texts that have been annotated with various additional information that will generally be used for various purposes in the field of linguistics is called a corpus. The development of the Indonesian national corpus has been mentioned by the Agency for the Assessment and Application of Technology, but so far no scientific publication has been found about it [20]. There are no language resource providers that can be accessed by the general public yet, while those available are provided by foreigners [26].

This study conducted a systematic literature study on the corpus of languages in Indonesia. The main purpose of this literature review process is to extract information from scientific articles related to the corpus of languages in Indonesia, such as the type of corpus used, the language used, methods and objectives of development, the purpose of using the corpus and the domain of the research.

2 Methods

Systematic literature review has three main stages, namely review planning, review process, and reporting the results [13]. This study refers to the existing

methods used in the literature review process. In this study, the review guidelines carried out by [14] have stages, Identification, Screening, Eligibility, and Included.

This literature review begins by getting articles according to the needs. At this stage, scientific articles are searched from the journal indexing database system. The journal database site used comes from national and international sources, nationally obtained from "Garba Rujukan Digital" (Garuda)[1]. Meanwhile, international sources were obtained from IEEExplore[2], ACM[3] and Springerlink[4]. The steps of the literature review process are carried out in the following Fig. 1.

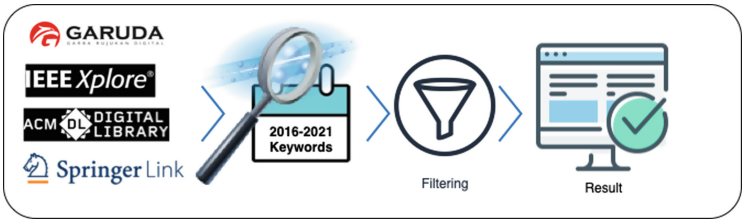


Fig. 1. Literature review process

The search was carried out on the abstract of published articles in the 2016–2021 range. The abstract must contain keywords such as "corpus Indonesian language" or "korpus Bahasa Indonesia". The screening or filtering process is carried out by eliminating duplication, selecting articles in the fields of computer science and linguistics, and discussing only regional languages or the Indonesian national language. Articles containing foreign language corpus are not included for further processing. Restrictions on the type of corpus are also limited to written corpus or text.

The eligibility and include process is conducted by selecting relevant articles for review. The selection of articles focused on articles related to the language corpus in Indonesia. The development and use of the corpus for various purposes, such as for text summarization, sentiment analysis or natural language are included in this process. Selected articles have one of the following criteria: mentioning the type of corpus, the purpose of using the corpus, the method or purpose of developing the corpus and the domain of the research conducted.

3 Result and Discussion

Extraction of scientific articles related to the corpus in Indonesia is the main objective of this literature review process. The results of searching scientific

[1] https://garuda.kemdikbud.go.id/.
[2] https://ieeexplore.ieee.org/Xplore/home.jsp.
[3] https://dl.acm.org.
[4] https://link.springer.com.

articles using predetermined keywords and filtering for articles published in the last 5 years (2016–2021) on each journal database site get results as shown in Fig. 2.

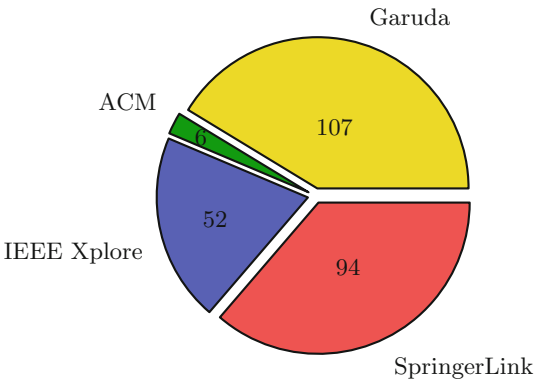


Fig. 2. Number of articles in each journal database

Two searches were carried out on the Garuda website with the keywords "korpus bahasa indonesia" and found a total of 55 documents, while as many as 62 documents were found containing the keywords "corpus Indonesian language". An abstract search containing three words namely corpus, Indonesian and language on the IEEE Xplore site yielded 52 documents and 6 documents on the ACM library. Searching for abstracts is not possible on the SpringerLink site and yielded 63 articles in the field of computer science and 31 articles in the field of linguistics. Searching on this site was carried out as shown in Fig. 3.

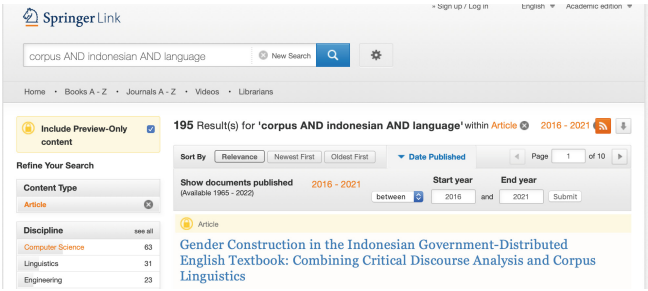


Fig. 3. Search on the SpringerLink

The screening and filtering process from Garuda database obtained 69 scientific articles that match the criteria. Two articles discusses the need for a language corpus in Indonesia in general, to save ethnic languages from extinction [11] and an exploration of the corpus of data on maritime local wisdom

texts [3]. Linguistic analysis on the corpus of Indonesian or local languages found 30 articles. Common uses of a corpus such as for emotion detection, typographical errors and others are found in 18 papers. A total of 19 papers discussed the use of the corpus for the translation of Indonesian and regional languages.

A total of 21 out of 52 articles after the screening and filtering process were obtained from search results through IEEE Xplore. There are 3 articles that discuss the development of the corpus in general, namely, the construction of thesaurus [23], NER (Named Entity Recognition) data set [1] and a special corpus of pornography [2]. From a total of 13 papers related to linguistic analysis, 5 papers contain the use of corpus in general. Six papers resulted from the ACM database. There was only 1 article met the criteria for the discussion and related to linguistic analysis, namely grammar correction [15]. From the SpringerLink database, there are 104 articles from the fields of computer science and linguistics, 3 of which are included in the literature review category. The three papers use the Indonesian corpus for topic search modeling [22], research on text classification accuracy [17] and examine two nominal prefixes in Indonesian [4].

Most of the papers used monolingual corpus, i.e. 19 articles and the rest used parallel or multilingual corpus. The use of raw corpus coming from the web, such as news sites, Wikipedia, or social media still dominates, namely 37 articles. A total of 19 papers used existing corpus such as Leipzig[5] or idn-treebank[6] or using data from previous studies. The development of a manual corpus from books or the like were also carried out on 20 papers and 18 papers did not clearly state the source of the corpus used. A summary of corpus types and sources is shown in Fig. 4.

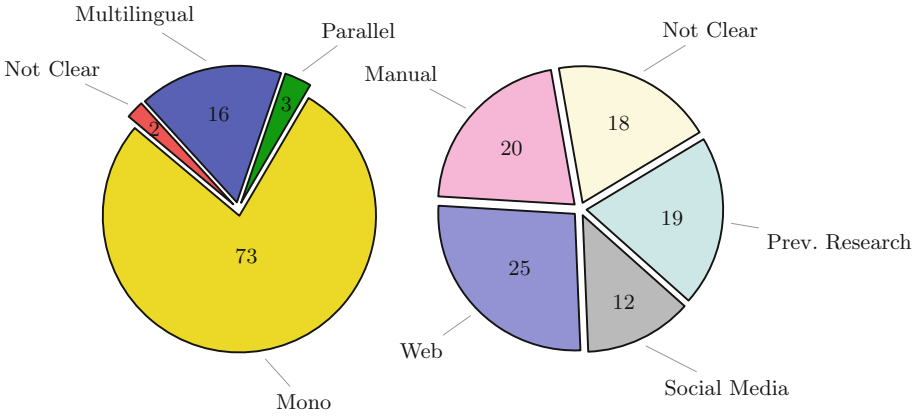


Fig. 4. Corpus types and sources

[5] https://corpora.uni-leipzig.de/en?corpusId=ind_mixed_2013.
[6] https://github.com/famrashel/idn-treebank.

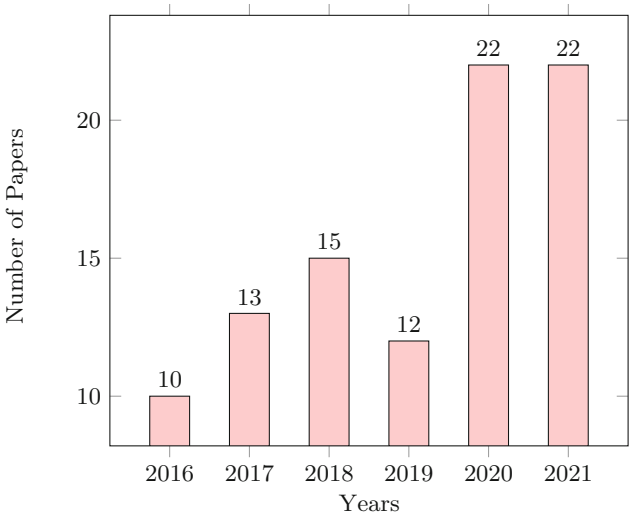


Fig. 5. Increasing number of articles

The results of the review paper also show an increasing trend in the number of studies related to the corpus of languages in Indonesia as shown at Fig. 5. In the 2020 and 2021 papers, it appears that the research topics are increasingly diverse and indicate the need for an annotated corpus. Such as research [27] which defines object and complement, Part of Speech [5,7,9], NER [21], meaning [18,31] and MWE (Multiword Expression) [10]. A total of 15 studies in 2019–2021 were related to the parallel corpus of regional-Indonesian languages.

4 Conclusion and Future Work

From the results obtained, most of the papers still focus on the use of the corpus, not on the development of the corpus of languages in Indonesia. Research papers obtained from the Garuda database contain interesting findings, namely an increase in the number of papers related to the translation of regional languages into Indonesian.

The corpus used for research materials in most of the papers utilizes data from primary sources such as social media, Wikipedia, or internet news sources and some using corpus that have been specially collected. The results of the literature review also show that only a few articles related to the development of the corpus are found in the Garuda database. None of articles discusses the development of the multilingual corpus, languages in Indonesia.

Thus, research for the development of a corpus of languages in Indonesia are still very wide open. The need for a multilingual corpus, languages in Indonesia is clearly seen from the results of the review. Our next research plan is to develop a framework for the collection, processing and storage of annotated multilingual corpus of languages in Indonesia.

References

1. Alfina, I., Manurung, R., Fanany, M.I.: DBpedia entities expansion in automatically building dataset for Indonesian NER. In: 2016 International Conference on Advanced Computer Science and Information Systems (ICACSIS), pp. 335–340. IEEE (2016)
2. Andriansyah, M., Purwanto, I., Subali, M., Sukowati, A.I., Samos, M., Akbar, A.: Developing Indonesian corpus of pornography using simple NLP-text mining (NTM) approach to support government anti-pornography program. In: 2017 Second International Conference on Informatics and Computing (ICIC), pp. 1–4. IEEE (2017)
3. Darmoko: Revitalisasi teks-teks kearifan lokal kemaritiman untuk membangun kehidupan bermasyarakat, berbangsa, dan bernegara. PUSTAKA: Jurnal Ilmu-Ilmu Budaya (2019)
4. Denistia, K., Baayen, R.H.: The Indonesian prefixes pe- and pen-: a study in productivity and allomorphy. Morphology **29**(3), 385–407 (2019)
5. Dewi, N.P., Ubaidi, U.: Pos tagging bahasa madura dengan menggunakan algoritma brill tagger. Jurnal Teknologi Informasi dan Ilmu Komputer **7**(6), 1121–1128 (2020)
6. Dharma, A.: Pembinaan dan pengembangan bahasa daerah. In: Language Maintenance and Shift, pp. 8–11 (2011)
7. Dharmawan, E., Sujaini, H., Muhardi, H.: Perbandingan nilai akurasi terhadap penggunaan part of speech set pada mesin penerjemah statistik. JUSTIN (Jurnal Sistem dan Teknologi Informasi) **8**(3), 250–256 (2020)
8. Embram, E.R.: Rancangan awal pengembangan bahan ajar bahasa sentani. Kibas Cenderawasih **15**(1), 117–132 (2019)
9. Firmansyah, I., Adikara, P.P., Adinugroho, S.: Klasifikasi kelas kata (part-of-speech tagging) untuk bahasa madura menggunakan algoritme viterbi. Jurnal Teknologi Informasi dan Ilmu Komputer **8**(5), 1039–1048 (2021)
10. Gunawan, D., Amalia, A., Charisma, I.: Automatic extraction of multiword expression candidates for Indonesian language. In: 2016 6th IEEE International Conference on Control System, Computing and Engineering (ICCSCE), pp. 304–309. IEEE (2016)
11. Ibrahim, G.A., Mayani, L.A.: Perencanaan bahasa di indonesia berbasis triglosia. Linguistik Indonesia **36**(2), 107–116 (2018)
12. Kemdikbud: Bahasa dan peta bahasa di indonesia (2019).https://petabahasa.kemdikbud.go.id/. Accessed 21 Dec 2021
13. Kitchenham, B.: Procedure for undertaking systematic reviews. Computer Science Department, Keele University (TRISE-0401) and National ICT Australia Ltd (0400011T. 1). Joint Technical report (2004)
14. Liberati, A., et al.: The Prisma statement for reporting systematic reviews and meta-analyses of studies that evaluate health care interventions: explanation and elaboration. J. Clin. Epidemiol. **62**(10), e1–e34 (2009)
15. Lin, N., Chen, B., Lin, X., Wattanachote, K., Jiang, S.: A framework for Indonesian grammar error correction. Trans. Asian Low-Resour. Lang. Inf. Process. **20**(4), 1–12 (2021)
16. Liu, B., et al.: Sentiment analysis and subjectivity. Handbook Nat. Lang. Process. **2**(2010), 627–666 (2010)
17. Mutiara, A.B., Wibowo, E.P., Santosa, P.I., et al.: Improving the accuracy of text classification using stemming method, a case of non-formal Indonesian conversation. J. Big Data **8**(1), 1–16 (2021)

18. Nugraha, D.S.: Makna-makna gramatikal konstruksi verba denominatif dalam bahasa indonesia. Bahasa dan Seni: Jurnal Bahasa, Sastra, Seni, dan Pengajarannya **49**(2), 224–239 (2021)
19. Putra, K.A.: Youth, technology and indigenous language revitalization in Indonesia. Ph.D. thesis, The University of Arizona (2018)
20. Raharjo, S., Wardoyo, R., Putra, A.E.: Desain korpus indonesia (tinjauan informatika). In: Konferensi Lingustik Tahunan Atma Jaya 13 (KOLITA 2013), pp. 351–355 (2015)
21. Rosidy, A.S., Akhriza, T.M., Husni, M.: Kombinasi metode ner-ocr untuk meningkatkan efisiensi pengambilan informasi di poster berbahasa indonesia. Jurnal Teknologi dan Sistem Komputer **8**(4), 263–269 (2020)
22. Setiawan, R., Budiharto, W., Kartowisastro, I.H., Prabowo, H.: Finding model through latent semantic approach to reveal the topic of discussion in discussion forum. Educ. Inf. Technol. **25**(1), 31–50 (2019). https://doi.org/10.1007/s10639-019-09901-7
23. Sholikah, R.W., Arifin, A.Z., Purwitasari, D., Fatichah, C.: Co-occurrence technique and dictionary based method for Indonesian thesaurus construction. In: 2017 5th International Conference on Information and Communication Technology (ICoIC7), pp. 1–6. IEEE (2017)
24. Simons, G.F.: Two centuries of spreading language loss. Proc. Linguist. Soc. Am. **4**(1), 1–27 (2019)
25. Strochlic, N.: The race to save the world's disappearing languages (2018). https://news.nationalgeographic.com/2018/04/saving-dying-disappearing-languages-wikitongues-culture. Accessed 10 Dec 2021
26. Suhardijanto, T., Dinakaramani, A.: Korpus beranotasi: Ke arah pengembangan korpus bahasa-bahasa di Indonesia. Prosiding Kongres Bahasa Indonesia. Prosiding Kongres Bahasa Indonesia, pp. 339–355 (2019)
27. Sutami, D.P.H.: Pendefinisian objek dan pelengkap dalam kamus besar bahasa indonesia sebagai penunjang literasi. Linguistik Indonesia, p. 179 (2020)
28. Tondo, H.: Kepunahan bahasa-bahasa daerah: faktor penyebab dan implikasi etnolinguistis. Jurnal masyarakat dan budaya **11**(2), 277–296 (2009)
29. Wicaksono, A.F., Vania, C., Distiawan, B., Adriani, M.: Automatically building a corpus for sentiment analysis on Indonesian tweets. In: Proceedings of the 28th Pacific Asia Conference on Language, Information and Computing, pp. 185–194 (2014)
30. Wijana, D.P.: Pemertahanan dan pengembangan bahasa indonesia (Indonesian language maintenance and development). Widyaparwa **46**(1), 91–98 (2018)
31. Yuki, L.K.: Study of the meaning of Lelaki, Bujang and Bujangan words in the utilisation of Indonesian corpus. Jurnal Sasindo UNPAM **9**(1), 81–93 (2021)